

Tropical Meteorology



Chapter 4: Monsoon



Objective

- *To Define Monsoon*
- *To Know The physical mechanisms leading to monsoon circulation*
- *To Identify the areas of monsoon circulation*
- *To know the Role of Monsoons in Tropical Climate*

Monsoons

Definition

- The name *monsoon* is derived from the Arabic word *mausim*, which means season, referring to large-scale seasonal reversals of the wind regime.
- The monsoons of the world are defined through the seasonal reversal of large-scale surface winds.
- It is a seasonal wind system driven by land–ocean heating contrast, resulting in wet summers and dry winters.

Conti....

- The best place to see an illustration of this phenomenon is in the domain of the Indian Monsoon where the southwesterly surface winds of the summer monsoon get replaced by the northeasterly surface winds of the winter monsoon.
- The important feature of the monsoon is the heavy precipitation associated with the monsoon onset.

Characteristic Features Of Monsoons

Characteristic Features Of Monsoons

- ❑ Seasonal Wind Reversal
- ❑ Distinct Wet and Dry Seasons
- ❑ Strong Rainfall Concentration
- ❑ Large-Scale Circulation System
- ❑ Linked with Pressure Systems

The physical mechanisms leading to monsoon circulation

Conti....

- The basic trigger for the monsoon is a contrast between the temperature of the Asian continent and that of the surrounding ocean.
- The thermal anomaly in the middle troposphere is enhanced by the spectacular topography of the Tibetan Plateau, which extends upward to about the 500 mb level.

Conti....

- Large areas of the tropics are influenced by monsoon circulations *which result from the temperature contrast that results from the different temperature responses to heating of land and ocean.*
- Warming of the land relative to the ocean leads to “onshore” flow, cumulus convection, and warming of the troposphere by latent heat release; the reverse circulation occurs during the winter.

Conti....

- Entire economies in parts of the tropics depend on an adequate supply of monsoon rains.
- Knowledge of monsoon variability, including the onset and retreat of the monsoon, breaks in the monsoon, and rainfall amounts is critical information for billions of people in Asia and Africa.
- Too little or too much rainfall can have devastating consequences.

Conti....

- *The basic drive for a monsoon circulation is provided by the contrast in the thermal properties of the land and sea surfaces.*

Conti.....

- The Indian Monsoon turns large parts of India from a kind of semi-desert into green lands.
- In places like this it is crucial for farmers to have the right timing for putting the seeds on the fields, as it is essential to use all the rain that is available for growing crops.

Effects Of The Tibetan Plateau On The Onset Of The Summer Monsoon

Conti....

- Much of the “surface” heating associated with the summer monsoon actually occurs in the middle troposphere, because it is located on the Tibetan Plateau.
- The Tibetan Plateau towers above the surrounding land surface, with average elevations over the central Plateau of over 3000 m.
- An island of warm air is centered over the Tibetan plateau.

Conti....

- The onset of the summer monsoon brings cooler surface temperatures to India and other areas that receive monsoonal precipitation, due to the increase in clouds as well as the increase in soil moisture that accompanies the precipitation.

Conti....

- From an agricultural standpoint, the beginning of the precipitation associated with the Asian summer monsoon is probably one of the most anticipated events in the world.
- The onset of the monsoon is generally defined as the beginning of consistent rainfall of the monsoon season.

Types of Monsoons

Based on Season

Summer

- Land heats faster than ocean
- Land → low pressure (heat low)
- Ocean → high pressure

Wind flows:

Ocean → Land

Moist air → rainfall

Winter

- Land cools faster than ocean
- Land → high pressure
- Ocean → low pressure

Wind flows:

Land → Ocean

Dry air → little/no rainfall

Based on Region

- Asian Monsoon
- African Monsoon
- Australian Monsoon

Areas Of Monsoon Circulation

Conti....

- Much of the tropics is influenced by monsoons. **The most extensive monsoon circulation by far is the Asian monsoon.**
- This monsoon completely dominates the climate of the Indian subcontinent, producing warm wet summers and cool dry winters.

Conti....

- Ramage (1971) defined a global monsoon domain based on the Asian, African and Australian monsoon systems.
- That domain is illustrated in Fig below This illustration schematically portrays the surface wind reversal between the Winter Monsoon (relative to the Northern Hemisphere) shown by solid lines, and its Summer counterpart, shown by dashed lines.

DOMAIN OF MONSOONS, REVERSAL OF SURFACE WIND DIRECTIONS

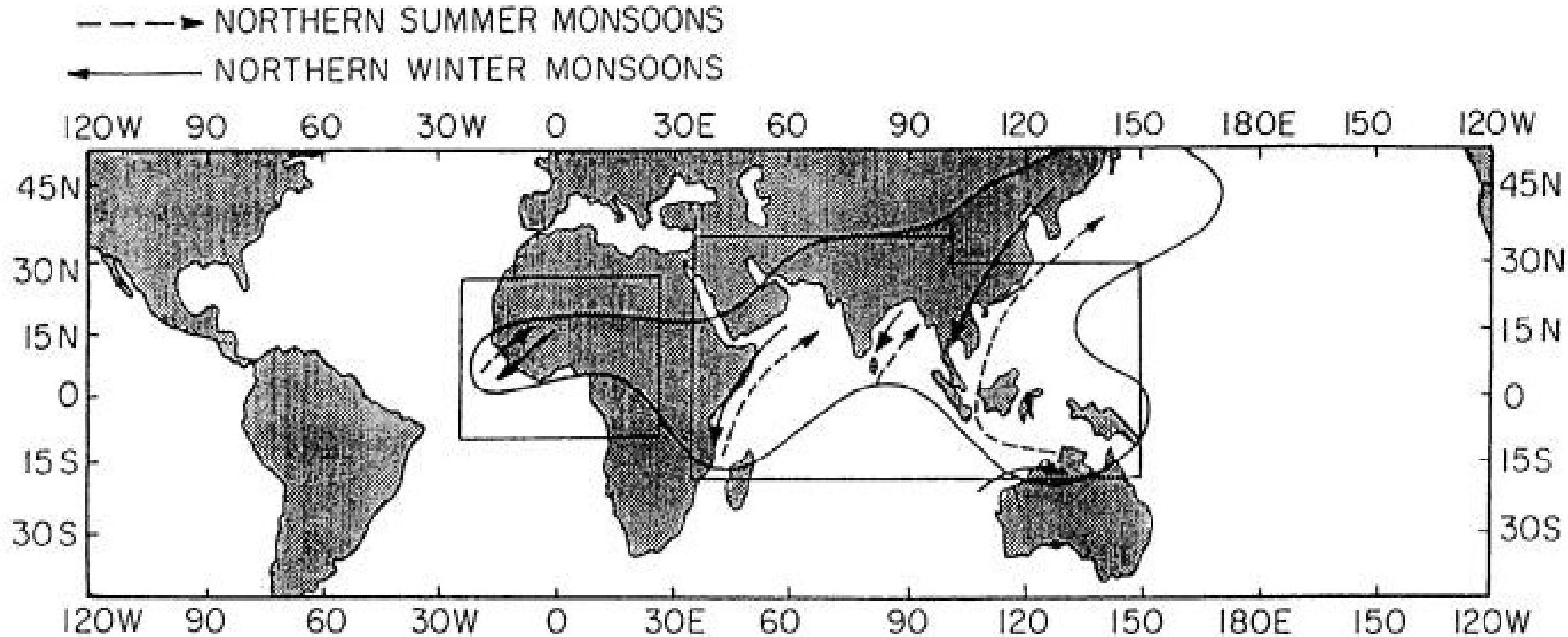


Fig. 5.2 Domain of monsoons (After Ramage 1971)

Conti....

- Monsoons occur in many parts of the world. They can be viewed as thermally forced stationary planetary waves associated with land-sea contrast.
- The most spectacular monsoon on Earth is the one associated with Asia, our largest continent.

Conti....

- The Asian monsoon can be defined in a number of ways. By definition, a monsoon is a dramatic seasonal reversal of the low-level prevailing winds.
- In the dramatic case of the Asian monsoon, the winds near the surface reverse from the northeast in winter to the southwest in summer

Conti....

- The 15 m s⁻¹ low-level southwest wind that crosses the Equator and flows from the coast of east Africa to the shores of India is known as the Somali jet. It is one of the strongest low-level jets in the world.
- There are other seasonal changes of wind in the world, but none have the geographical scope or the socioeconomic impact of the Asian monsoon.

Monsoons around the world

- While the Asian monsoon is the most widely known, monsoon conditions also occur also (though to a lesser degree) in northern Australia, parts of western, southern and eastern Africa, and parts of North and South America.

The African monsoon

- The monsoon of western Sub-Saharan Africa is the result of the seasonal shifts of the Intertropical Convergence Zone and the great seasonal temperature and humidity differences between the Sahara and the equatorial Atlantic Ocean.
- The ITCZ migrates northward from the equatorial Atlantic in February, reaches western Africa on or near June 22, then moves back to the south by October.

Conti....

- The main two components of the African Monsoons are the West African Monsoon, which prevails during the Northern Hemisphere Summer (June through September), and the East African Monsoon with rains during spring (MAM) and autumn (OND).
- The precipitation and atmospheric circulation patterns are shown in the adjacent figure. Combined influence of the Indo-Pacific and the Atlantic Oceans drive the inter-annual and the decadal monsoon variability over these regions.

Conti.....

- The key features of the West African Monsoon are the low level southwesterly flow from the Atlantic Ocean and the Inter-Tropical Convergence Zone (ITCZ) north of the equator.
- Monsoon rainfall over West Africa occurs during the June through September period (Figure left panel). The West African Sahel is well known for the severe droughts that ravaged this region during the 1970s and 1980s.
- The vagaries of the rainfall in Sub-Saharan Africa have a profound and often dire consequence on African society and economy.

Conti....

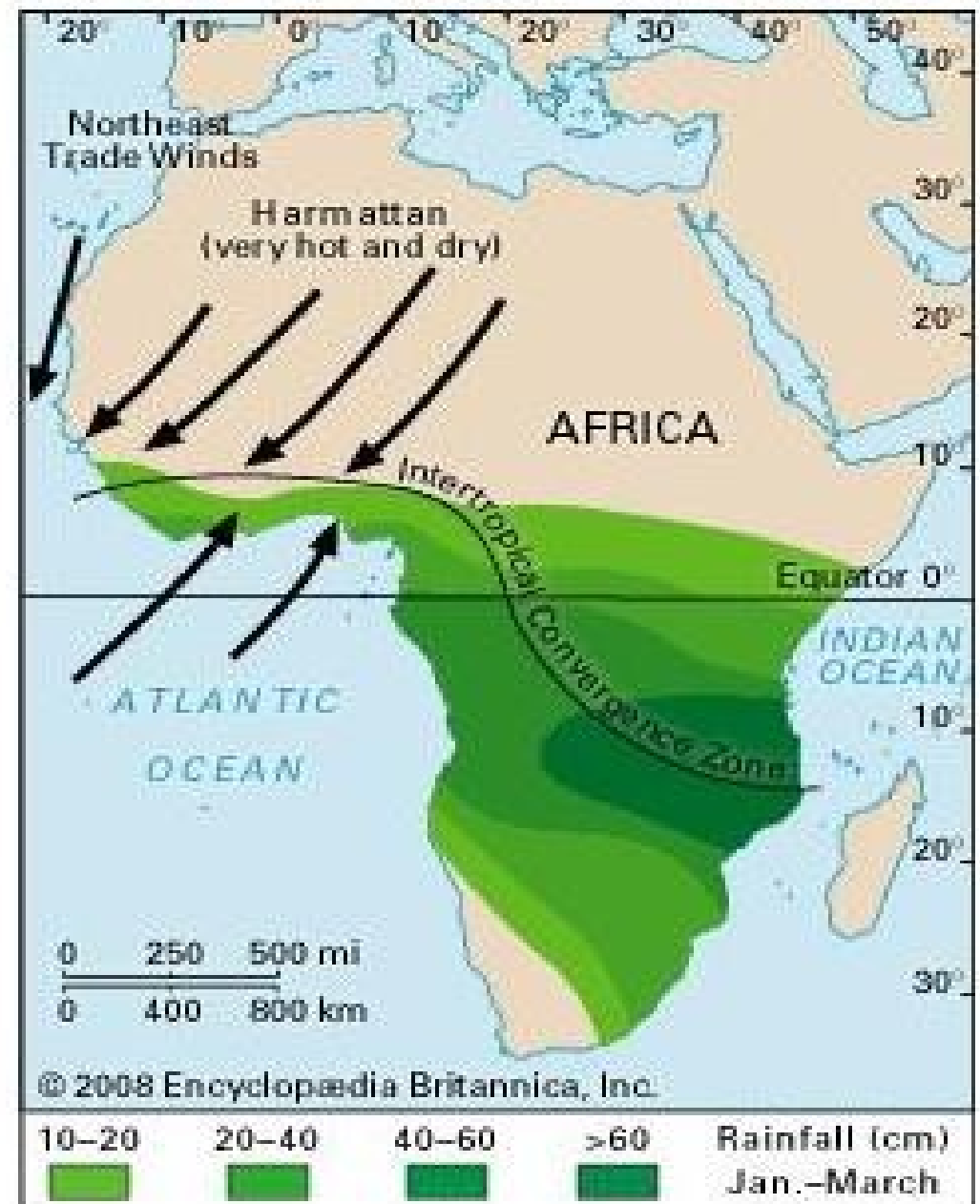
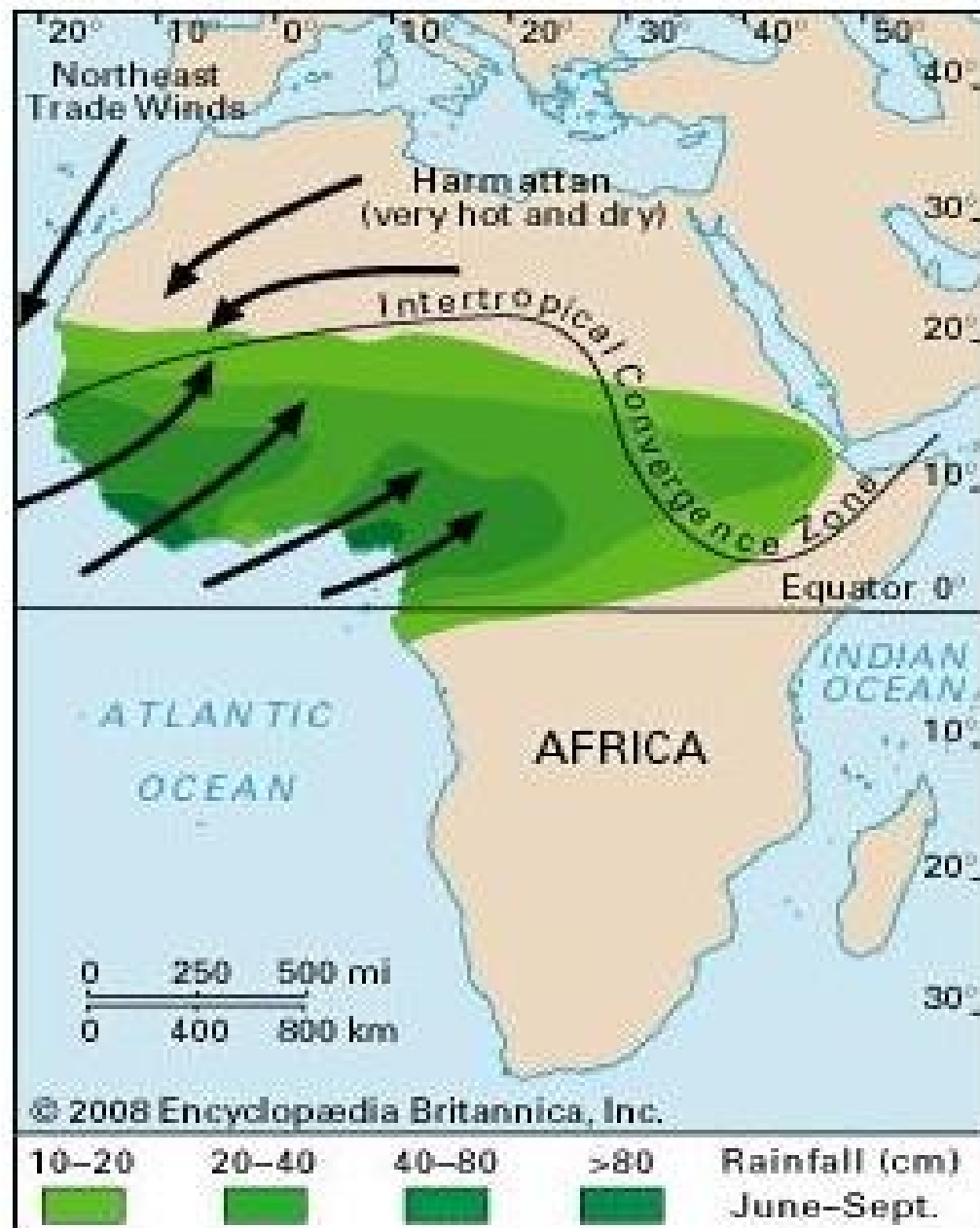
- The East African Monsoon is associated with the ITCZ moving south of the equator.
- The so-called long rains prevail during spring (MAM) and the short rains during autumn (OND).
- Thus the transition (equinox) seasons bring most rainfall to East Africa (Figure right panel).
- Abrupt decline in the long rains have resulted in droughts over this region.

Conti.....

- To reduce the vulnerability of the African community to the variations in the strengths of the two monsoons, scientific community needs to improve the reliability of forecasts to enable forward planning.
- Understanding the monsoon systems and their predictability over Africa has highly improved through AMMA (African Monsoon Multidisciplinary Analysis), the main international program on climate and environment in Africa.
- AMMA is planned to continue research for a further decade focused on the integration of weather and climate science with decision making for the African society.

Conti....

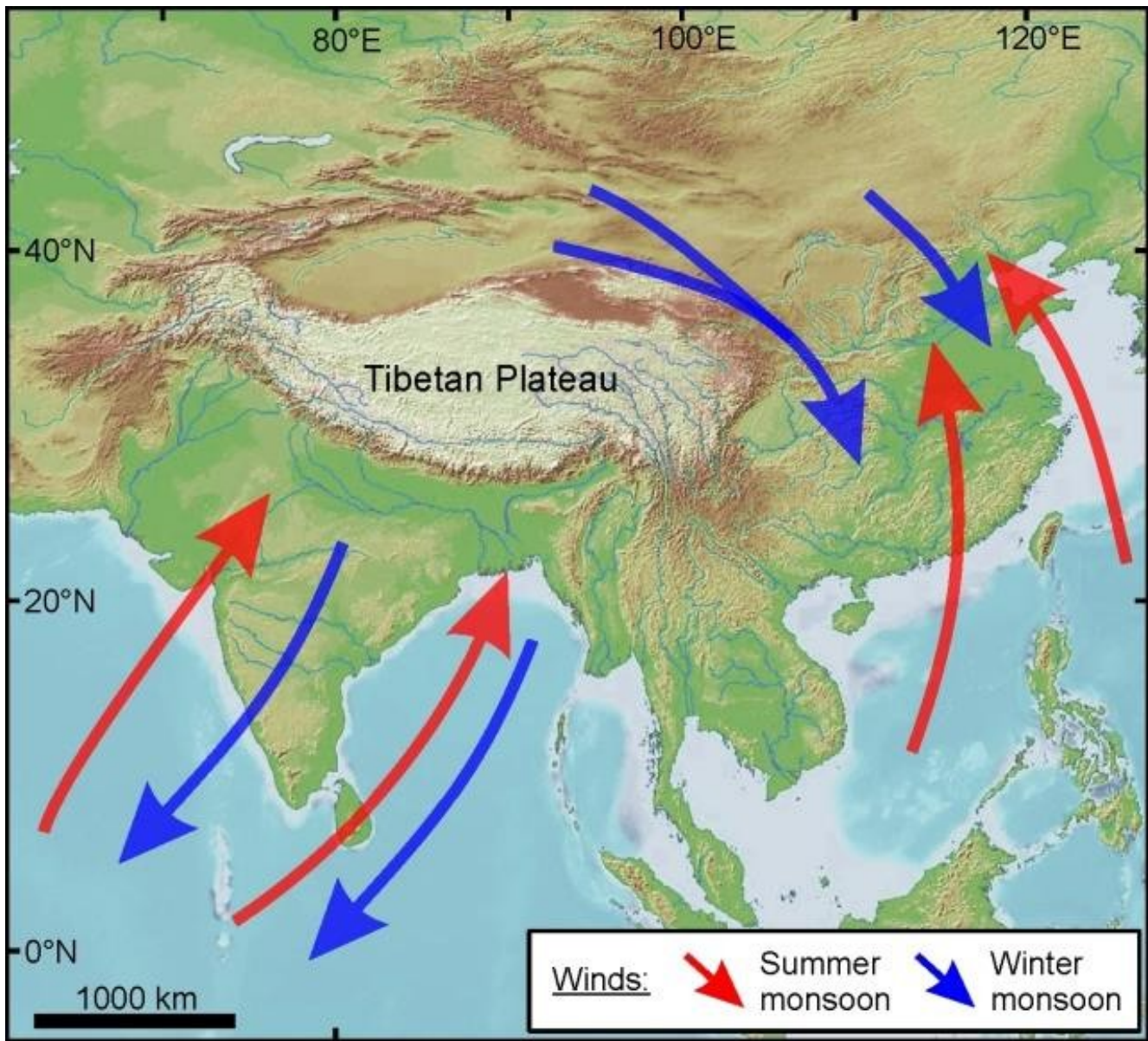
- The major difference between the circulations of the two regions(Asian and African) is due largely to the differing geography of the land to sea distribution and to the lack of a large mountain range to the north of West Africa.



Asian monsoons

Conti....

- The Asian monsoons may be classified into a few sub-systems, such as
- The *Indian Subcontinental Monsoon* which affects the Indian subcontinent and surrounding regions including Nepal.
- The East *Asian Monsoon* which affects southern China, Taiwan, Korea and parts of Japan.



Southwest monsoon



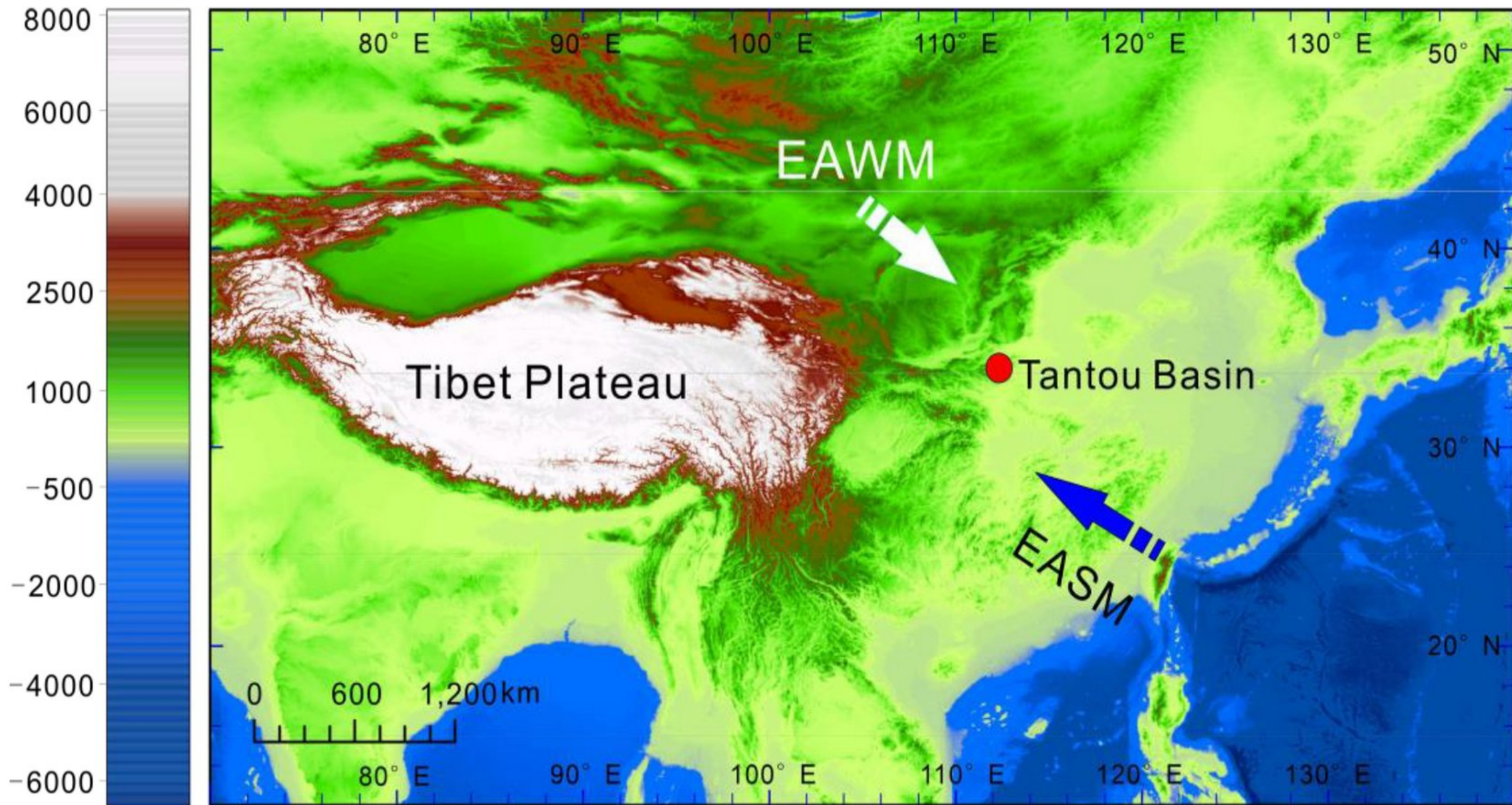
Conti....

- The monsoon accounts for nearly 80% of the rainfall in India.
- Indian agriculture (which accounts for 25% of the GDP and employs 70% of the population) is heavily dependent on the rains, for growing crops especially like cotton, rice, oilseeds and coarse grains.
- A delay of a few days in the arrival of the monsoon can badly affect the economy, as evidenced in the numerous droughts in India in the 1990s.
- Indian monsoon is due to Temperature differences between the Himalayas and the Indian Ocean.

East Asian Monsoon

- The East Asian monsoon affects large parts of Indochina, the Philippines, China, Taiwan, Korea and Japan.
- It is characterised by a warm, rainy summer monsoon and a cold, dry winter monsoon.
- The rain occurs in a concentrated belt that stretches east–west except in East China where it is tilted east-northeast over Korea and Japan.
- The seasonal rain associated with the East Asian monsoon is known as Meiyu in China, Jangma in Korea, and Baiu in Japan

- The onset of the summer monsoon is marked by a period of pre monsoonal rain over South China and Taiwan in early May.
- From May through August, the summer monsoon shifts through a series of dry and rainy phases as the rain belt moves northward, beginning over Indochina and the South China Sea (May), to the Yangtze River Basin and Japan (June) and finally to northern China and Korea (July).
- When the monsoon ends in August, the rain belt moves back to southern China.



Asian Monsoon Characteristic Features

- A monsoon climate is characterised by a dramatic seasonal change in direction of the prevailing winds of a region which brings a marked change in rainfall. The monsoon climate results in high annual rainfall totals exceeding 1.5 m (5 ft) in many places.
- Monsoons lead to distinct wet and dry seasons in many areas throughout the tropics and are most often associated with the Indian Ocean.

Australian Monsoon

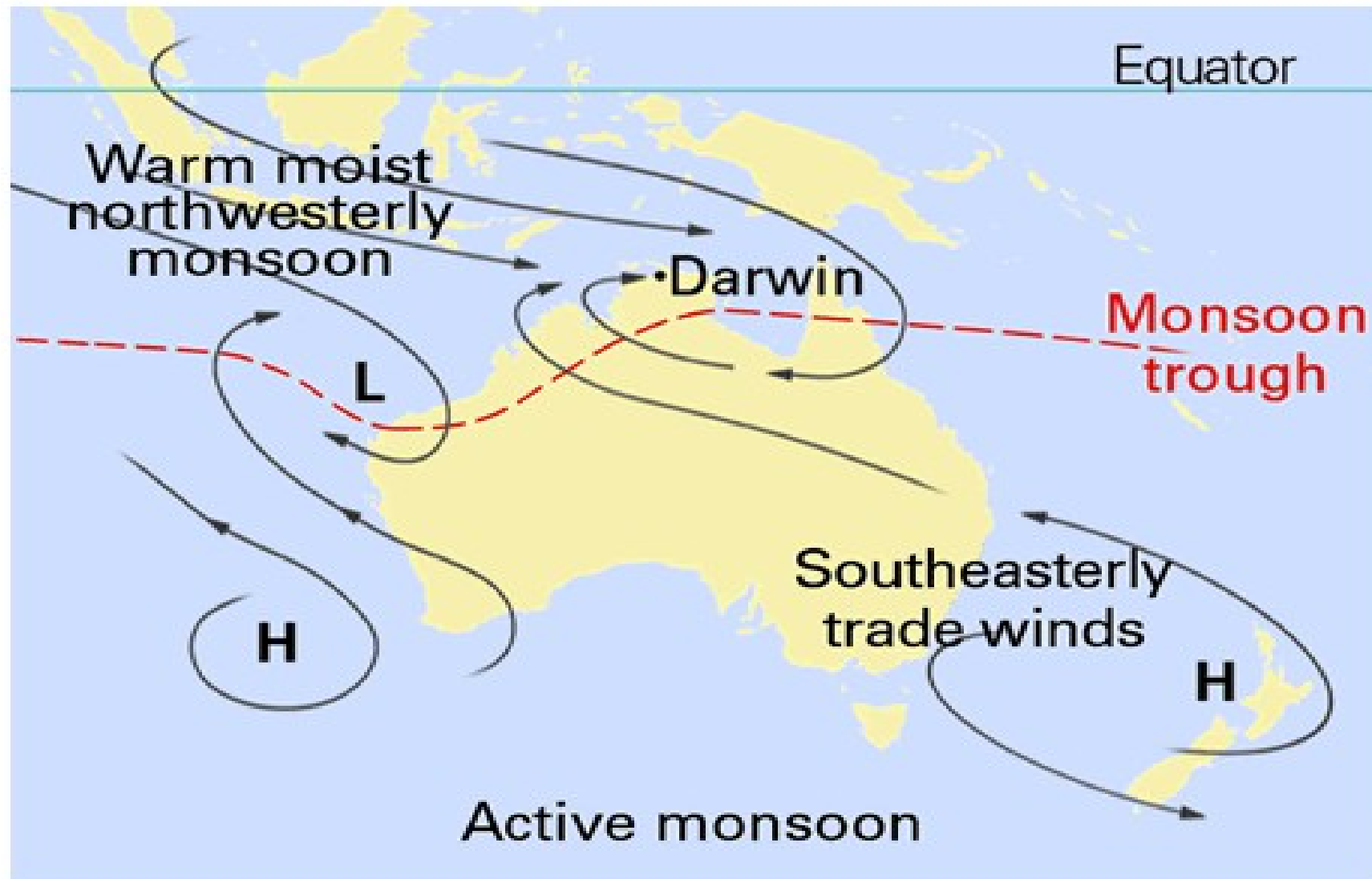
Conti....

- Australia is due to Temperature differences between the Australian landmass and surrounding oceans
- In northern Australia, the prevailing wind is from the east or **southeast** for most of the year, but during active monsoon periods (occurring any time during **November to April**) the winds shift to become *northwesterly* at the surface As the Australian summer approaches, the continent heats up.
- Low pressure is created, which effectively draws the monsoon trough - a zone of low pressure and rising air – over northern Australia.

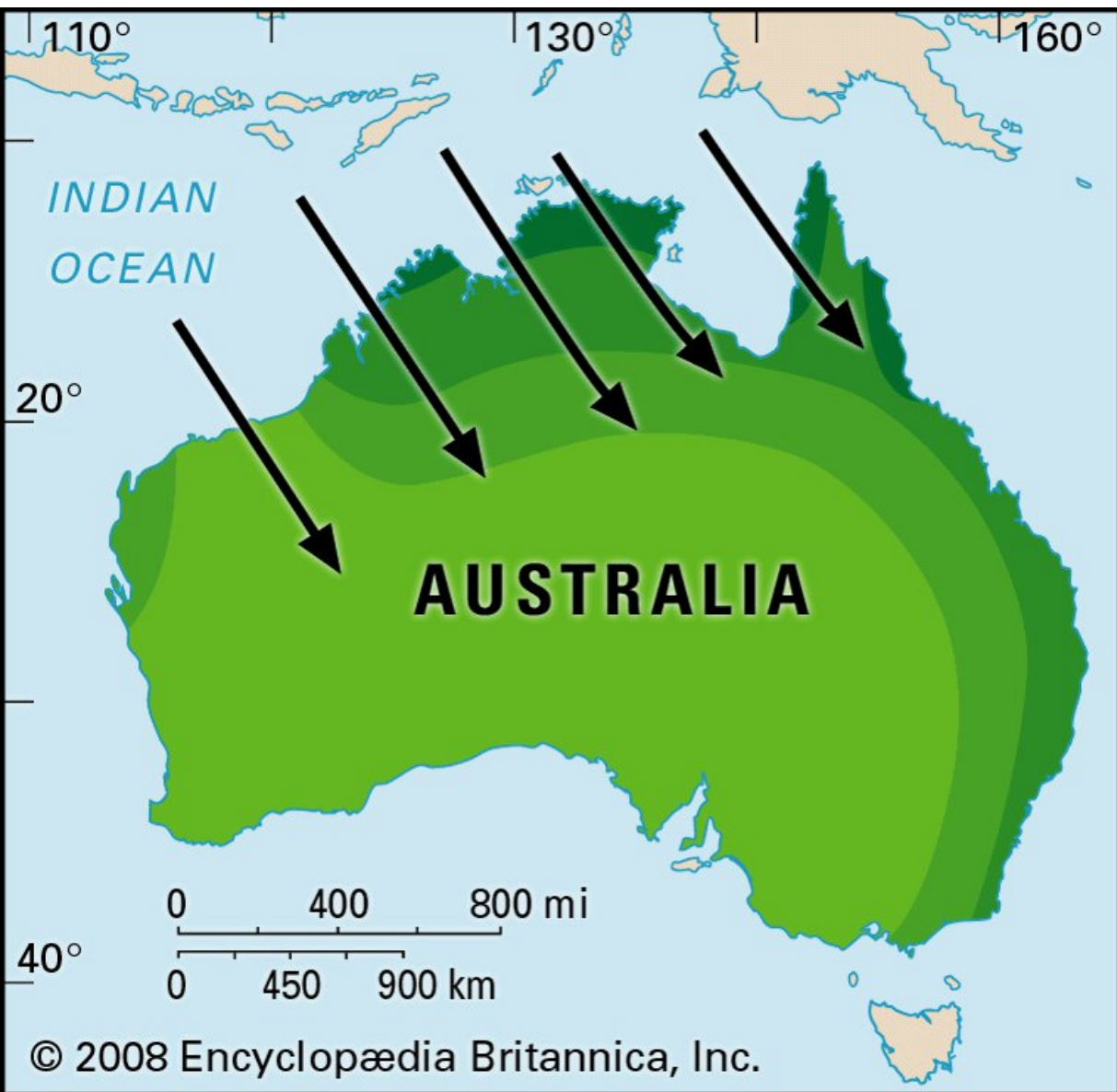
Conti....

- This trough draws in moist air from the surrounding oceans and this influx of moist air is referred to as the monsoon.
- The monsoon can be in either an "active" or an "inactive" phase.
- The **active phase** is usually associated with broad areas of **cloud and rain**, with sustained moderate to fresh northwesterly winds on the north side of the trough.

- Widespread heavy rainfall can result if the trough is close to, or over, land. An inactive or "break" period occurs when the monsoon trough temporarily weakens or retreats north of Australia.
- It is characterised by light winds and isolated shower and thunderstorm activity, sometimes with gusty squall lines.
- Transitions from active to inactive monsoon phases may be associated with the Madden-Julian Oscillation, a large-scale slow-moving band of increased cloudiness that travels eastwards in the tropics.



The diagram above shows the synoptic patterns typically associated with the active phase of the north Australian monsoon. The dashed red line indicates the position of the monsoon trough, which will shift slightly as weather systems progress across the continent. The monsoon trough is an area of low pressure along which broad-scale convergence and convection occurs.



How does it affect Australia?

- The monsoon is associated with **cloudy conditions**, lengthy periods of heavy rain, occasional thunderstorms and fresh to strong squally winds. This often causes flooding in affected areas.
- Over **three-quarters of annual rainfall in** Northern Australia falls during this time.

The Role of Monsoons in Tropical Climate

Role of Monsoons in Tropical Climate

- ❑ They Determine The Wet And Dry Seasons
- ❑ Supporting or Influencing Agriculture
- ❑ Regulates Temperature
- ❑ Drives Atmospheric Circulation
- ❑ Influences Weather Systems
- ❑ Impacts Society

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Thanks !